Agentic Coding for Economists

Five-hour hands-on workshop

Participant repo: `https://github.com/meleantonio/AC4E_Padova`

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How We Teach Today

Demo-first rhythm

1. **Short demo ($\leq$8 min):** One tactical workflow, live.
2. **Hands-on sprint:** You produce a file in *your* `examples/starter_article/` copy.
3. **Debrief:** What did the agent do vs what did you sign off?

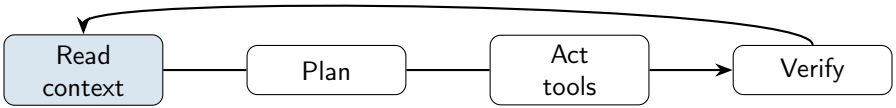
What is Agentic Coding?

Definition

Agentic coding = delegating implementation work to AI agents that **read**, **plan**, **edit**, and **verify**—while you orchestrate scope, context, and quality gates.

- **Agents** operate on your codebase with *tools*: edit files, run terminal commands, search the repo (and optionally the web).
- **You** control what they see (context files, attachments, web resources) and how they behave (the harness: `AGENTS.md`, rules, skills, hooks, ...).
- **Output** must be *verifiable*: tests pass, replication is valid, acceptance criteria achieved.

The Agent Loop



- **Read** — Project brief, `AGENTS.md`, attached files, search results. Bad context ⇒ bad code/output.
- **Plan** — Break work into steps (Plan mode, specifications, requirements, tasks, explicit goal).
- **Act** — Edit code, run code/tests, execute terminal commands, update paper write up.
- **Verify** — Re-run scripts, check replication pipeline, etc.

Why Economists Need This

- **Pipeline complexity:** Data → cleaning → estimation → tables → paper—dozens of small, automatable steps.
- **Reproducibility pressure:** Journals and replicators expect documented data, portable paths, one-command runs.
- **Multi-artifact projects:** Code, LaTeX, appendix, and replication package must stay consistent.
- **This workshop:** Build a *harness* you can fork onto a research project, not just one-off chat prompts.

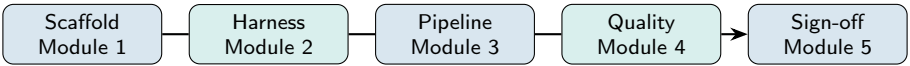
What is a Research Article Harness?

Harness

A **project template** where context files, skills, subagents, hooks, and verification gates work together across the research lifecycle, from idea to replication.

- **Not** a single magic prompt.
- **Is** repeatable infrastructure: the agent always knows the repo map, boundaries, and how to run checks.
- **Can** be adapted to other research projects.

Harness Lifecycle (Workshop Spine)



- **Scaffold** — Create repo, project idea/brief, `AGENTS.md`, privacy boundaries, three tasks.
- **Harness** — Create skills, subagents, hooks, MCPs; run one long goal with stop conditions.
- **Pipeline** — Update idea/brief, write and maintain code and paper.
- **Quality** — Referee report, replication check
- **Sign-off** — Checklist + 7-day adoption plan for your own research.

Demo: Coding Agent Interface

A typical coding agent UI:

- **Prompt/Chat:** Type a goal
  (e.g., "Clean `data.csv` and compute summary stats.")
- **Context** loaded: repo map, `AGENTS.md`, `templates/project_brief.md`.
- **Artifacts:** View/edit code, instructions, logs, or generated files.
- **Tools:** Run code, preview plots, test outputs (e.g., buttons for "Run All", "Test", "View PDF").

Project Context: AGENTS.md

What it is

The "front door" for any agent: a human-readable brief at the repo root that loads every session.

Typically includes:

- **Research question** and stage (seminar draft, JMP, etc.)
- **Repository map** — which folders agents may read or write
- **Verification commands** — `pytest`, `src/estimate_did.py`, `latexmk`
- **Do-not rules** — e.g. never edit `data/raw/` without permission

Claude lane: may also use `CLAUDE.md`; keep `AGENTS.md` for collaborators.

Example: AGENTS.md (HANK Macro)

Same pattern for applied micro, structural, or macro repos.

```
# AGENTS.md - HANK macro (illustrative)

## Project
HANK: monetary policy with heterogeneous households; seminar draft.

## Repo map
Read: src/, models/, data/processed/, docs/
Write: src/, models/, docs/drafts/, output/
Do NOT: data/raw/, output/confidential/, secrets/

## Verify
pytest src/ | python models/run_HANK.py | latexmk -pdf docs/paper.
   tex

## Rules
No raw-data edits without permission; no API keys or microdata in
   context.
Tests must pass; document model changes in docs/; use a branch for
   review.
```

Exercise: Create Your Own `AGENTS.md`

Exercise: Draft an `AGENTS.md` for your own project.

Include the following sections

- **Project:** Briefly state your project's research topic and stage.
- **Repo map:** *Which folders should your agent read or write?* What should be off-limits?
- **Verify:** Specify at least one command to check code or outputs (e.g., `pytest`, `make`).
- **Rules:** Any do-not rules, data boundaries, or requirements for contributors/agents?

Tip: Imagine onboarding a new research assistant (human or AI)—what context and boundaries do they need on their first day?

(Start from `templates/AGENTS_economics.md` or the HANK macro example above.)

Research Spec: Brief and Design Memo

Project Brief:
`templates/project_brief.md`

- **What:** Scope, audience, in/out of scope
- **Why:** Stops agents from "helpfully" expanding the paper
- **When:** At the beginning of the project, Module 1 — personalize or keep demo

Example:

`examples/starter_article/docs/project_brief.md`

Design Memo:
`templates/research_design_memo.md`

- **What:** Hypotheses, data, identification, risks/threats
- **Why:** Plan before code; hub for SDD-lite
- **When:** Beginning of the project, but maintained throughout, Module 3 — identification section is human-owned

Example:

`examples/starter_article/docs/research_design_memo.m`

Spec-Driven Development:
`examples/starter_article/spec/intent.md` — optional; for ambitious projects (intent → requirements → design → tasks).

Exercise: Create Your Own Project Brief and Design Memo

Exercise: Draft a project brief and design memo for your own project. Start from `templates/project_brief.md` and `templates/research_design_memo.md`.

Project Brief

- **What:** Scope, idea,audience, in/out of scope

Design Memo

- **What:** Hypotheses, data, identification, risks/threats

Privacy Boundary: `.cursorignore` or `AGENTS.md`

What it does: Controls what the agent *indexes* and may send as context.

- **Privacy:** API keys, microdata, confidential drafts
- **Performance:** Large CSVs, PDFs, `.venv`

Different from `.gitignore`:
`.gitignore` = version control;
`.cursorignore` = AI context.

```
data/raw/confidential/
.env
*.dta
*_confidential*
```

What Are Agent Skills?

Skills extend the agent with **reusable workflows** stored as `SKILL.md` files.

- **Trigger:** The skill's `description` in YAML frontmatter matches your request.
- **vs subagents:** Skills augment the *main* agent—no separate context window.
- **vs rules:** Rules are always-on constraints; skills are procedures you invoke.

Exercise: Create Your Own Agent Skills

Exercise: Create an agent skill for your own project.

Skill

- **What:** Description, trigger, steps

What Are Subagents?

Subagents are specialized assistants with their own **context window** and prompt.

- **Why:** Context isolation—a reviewer should not share the same "implementation bias" as the coder.
- **Format:** `examples/starter_article/.cursor/agents/*.md` or `examples/claude/.claude/agents/*.md` with YAML frontmatter + role prompt.
- **Invoke:** "Use identification-reviewer on the design memo only" or tool-specific `/agents`.

Exercise: Create Your Own Subagent

Exercise: Create a subagent for your own project.

Subagent

- **What:** Description, trigger, steps

What Are Hooks?

Hooks run **deterministic** scripts at lifecycle events (session start, before/after tool use).

- **Why:** LLMs forget; hooks enforce reminders (privacy, no `rm -rf`, run tests after edits).
- **Where:** `examples/hooks/.codex/`, Claude `settings.json`, `examples/starter_article/.cursor/hooks.json`.
- **Trust:** Codex `/hooks`; Claude settings UI—you approve what runs.

Economics example: After editing `examples/starter_article/src/`, hook reminds: "Re-run `pytest` and replication-checker before claiming results."

Long-Running Goals and Checkpoints

Problem: One chat cannot hold a whole pipeline; context drifts.
Solution: Explicit goals with **allowed paths**, **forbidden paths**, and **stop conditions**.

Tool	Mechanism
Codex	`/goal` or goal UI; `codex resume`
Claude Code	Checkpoints; `/goal`; `claude -continue` (CLI only)
Cursor	Background agents; Agent checkpoints

Exercise: Write a Long-Running Goal

Exercise: Draft a long-running goal for your project using the following template.

Long-Running Goal Template

- **Goal:** (Describe your end goal or desired outcome clearly and concretely.)
- **Allowed paths:** (List productive actions the agent can take to achieve the goal.)
- **Forbidden paths:** (List actions or areas that are off-limits; e.g., avoid modifying published data, do not email external parties, etc.)
- **Stop conditions:** (Define clear stopping criteria, e.g., when the goal is achieved, a report is generated, all code passes tests, etc.)

Tip: Be explicit about boundaries and what success looks like. This helps both you and your agent stay aligned over time. Template: `templates/long_running_goal_prompt.md`

MCP Servers

MCP = Model Context Protocol — a standard to connect agents to **external tools**.

- **What:** Cursor (or other hosts) invoke MCP servers; the agent sees structured tools (e.g. FRED API).
- **Config:** `examples/starter_article/.cursor/mcp.json.example` → `.cursor/mcp.json`; API keys via environment variables, never in git.
- **When it helps:** Recurring macro series pulls, scoped filesystem access.
- **When to skip:** One-off downloads; workshop runs fine without FRED.

Verify MCP setup in your Cursor version. Docs: https://cursor.com/docs/context/mcp

Skills vs Subagents vs Main Agent

Use	When
Skill	Single-purpose procedure (replication-check, referee pass, SDD plan)
Subagent	Isolated review role; parallel second opinion on memo or PR
Main agent	Orchestration, merging edits, deciding scope

Invoke explicitly: "Run replication-checker on repo root" beats "check if this replicates."

Swarms: GitHub as Control Plane

A **swarm** coordinates several agents toward one project outcome — not one giant prompt.

- **Issues** = units of work with acceptance criteria and allowed files.
- **Labels:** `parallel`, `sequential`, `blocked-by:#N`, `ready-for-review`.
- **Rule:** independent tasks → parallel agents; dependent tasks → sequential launches after prior PR merges.
- **Planner** opens issues; implementers work branches; reviewers stay read-only.

Workshop plan: `orchestration/card_krueger_swarm.md`; issue template: `orchestration/cloud_agent_issue_template.md`

Example: Card-Krueger Swarm Streams

Stream	Label	Depends on	Reviewer
Data map audit	sequential start	none	`data-reviewer`
Cleaning and validation	sequential	data map	`replication-verifier`
Baseline estimate	sequential	cleaning	`replication-verifier`
Source note	parallel	source URLs only	`identification-reviewer`
Teaching write-up	sequential	baseline	`pr-reviewer`

- Record handoffs in `notes/orchestration_log_template.md`.
- Merge only after review; keep `main` stable.
- Neither swarms nor loops replace human merge approval.

Loop on Verification

Three phases, repeated until exit or escalation:

1. **Implement** — narrowly scoped change against acceptance criteria.
2. **Evaluate** — `loop-verifier` subagent or replication-checker skill.
3. **Revise** — address only listed blockers; record iteration in the orchestration log.

Verdicts: GREEN → human review; YELLOW → loop again; RED or 3 iterations → stop and open a new issue. Protocol: `orchestration/verification_loop.md`; skill: `examples/starter_article/.cursor/skills/loop-on-verification/`; evaluator: `examples/starter_article/.cursor/agents/loop-verifier.md`

Loop vs Swarm

Swarm

- Many issues in parallel waves
- GitHub labels and dependencies
- Merge after subagent review
- Example: `orchestration/card_krueger_swarm.md`

Verification loop

- One issue, one agent stream
- Sequential implement–evaluate cycles
- Exit when criteria are green
- Skill: `loop-on-verification`

Both require acceptance criteria first. Neither replaces human merge approval.

Goal Files and the `/goal` Command

A **goal** is persistent task state the agent retrieves across sessions — the local equivalent of a GitHub issue.

Field	Write verifiable criteria
name	one-line task identifier
description	economic output when done
approach	files to edit, method to use
acceptance_criteria	checkbox list; each item checkable by command
constraints	raw data untouched; synthetic caveat required

Worked examples: `orchestration/goals/`; template: `templates/long_running_goal_prompt.md`.

Attach via Claude Code `/goal`, Codex Goals UI, or Cursor Cloud Agent task description.

Harness Components (Summary)

Component	What it does (Cursor paths)
Project context	Always-on: `AGENTS.md`, `templates/AGENTS_economics.md`, `examples/starter_article/.cursor/rules/*.mdc`
Research spec	Plan before code: `templates/project_brief.md`, `templates/research_design_memo.md`, `examples/starter_article/spec/intent.md`
Skills	Procedures: `examples/starter_article/.cursor/skills/referee-checklist/`, etc.
Subagents	Isolated reviewers: `examples/starter_article/.cursor/agents/`
Hooks	Deterministic guardrails: `examples/hooks/.codex/`, `examples/starter_article/.cursor/hooks.json`
MCP	Optional external APIs: `examples/starter_article/.cursor/mcp.json.example`
Gates	Evidence: `examples/starter_article/replication/README.md`, `templates/referee_report.md`, `templates/review_log.md`
Orchestration log	Lightweight record: what you delegated vs reviewed (`notes/orchestration_log_template.md`)